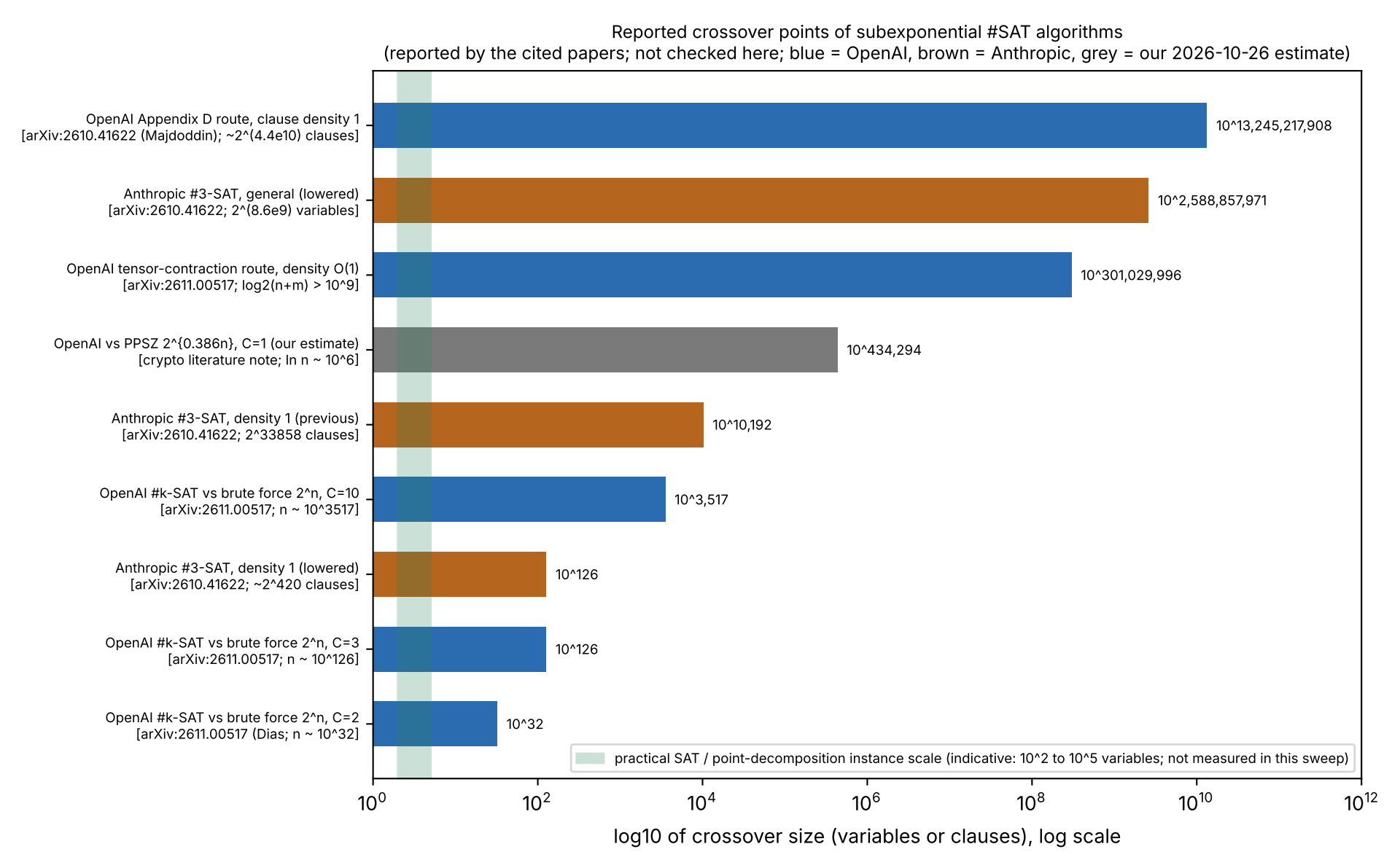
Reported crossover points of subexponential #SAT algorithms
(reported by the cited papers; not checked here; blue = OpenAI, brown = Anthropic, grey = our 2026-10-26 estimate)
OpenAI Appendix D route, clause density 1
[arXiv:2610.41622 (Majdoddin); ~2^(4.4e10) clauses]
10^13,245,217,908
Anthropic #3-SAT, general (lowered)
[arXiv:2610.41622; 2^(8.6e9) variables]
10^2,588,857,971
OpenAI tensor-contraction route, density O(1)
[arXiv:2611.00517; log2(n+m) > 10^9]
10^301,029,996
OpenAI vs PPSZ 2^{0.386n}, C=1 (our estimate)
[crypto literature note; ln n ~ 10^6]
10^434,294
Anthropic #3-SAT, density 1 (previous)
[arXiv:2610.41622; 2^33858 clauses]
10^10,192
OpenAI #k-SAT vs brute force 2^n, C=10
[arXiv:2611.00517; n ~ 10^3517]
10^3,517
Anthropic #3-SAT, density 1 (lowered)
[arXiv:2610.41622; ~2^420 clauses]
10^126
OpenAI #k-SAT vs brute force 2^n, C=3
[arXiv:2611.00517; n ~ 10^126]
10^126
OpenAI #k-SAT vs brute force 2^n, C=2
[arXiv:2611.00517 (Dias; n ~ 10^32]
10^32
practical SAT / point-decomposition instance scale (indicative: 10^2 to 10^5 variables; not measured in this sweep)
10^0
10^2
10^4
10^6
10^8
10^10
10^12
log10 of crossover size (variables or clauses), log scale